

Can we trust this person?



KYLIAN MBAPPÉ

FORWARD · WORLD CUP WINNER

Can we trust this person?



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On his own, nothing looks suspicious.

Can we trust this person?

PITCH STAFF



*ordered to
mop the pitch*



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FORWARD · WORLD CUP WINNER

On his own, nothing looks suspicious.

Can we trust this person?



On his own, nothing looks suspicious.

Can we trust this person?



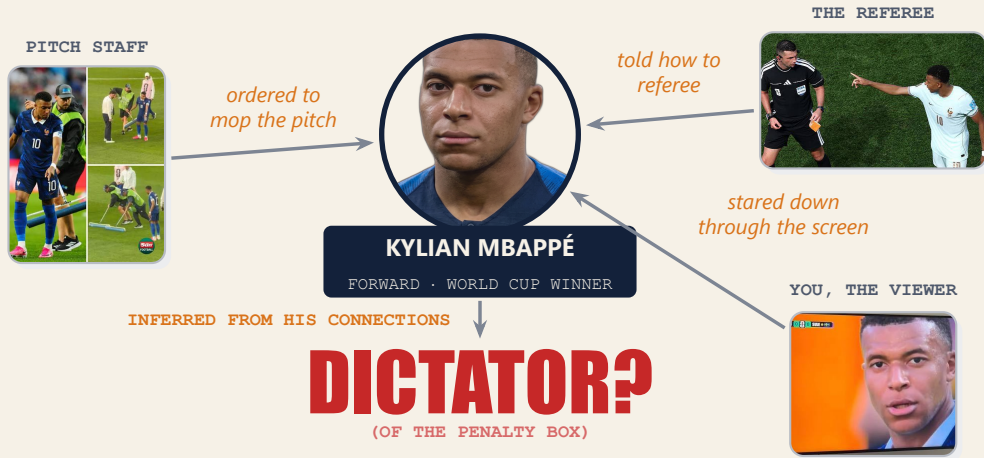
On his own, nothing looks suspicious.

Can we trust this person?



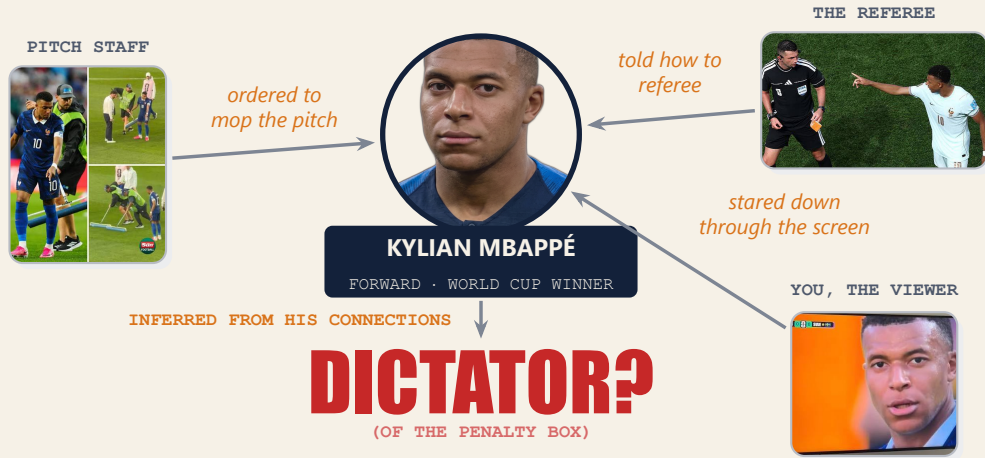
Wait... why is everyone taking orders from this guy?

Can we trust this person?



Wait... why is everyone taking orders from this guy?

What if the person itself isn't the information?



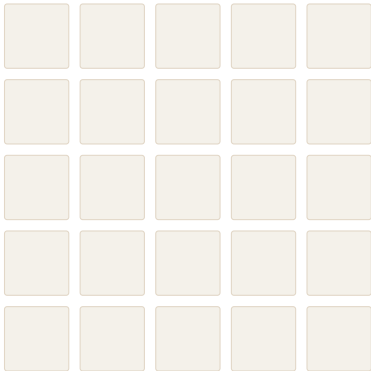
Wait... why is everyone taking orders from this guy?

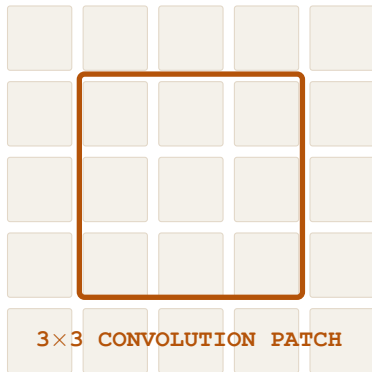
What if the person itself isn't the information?

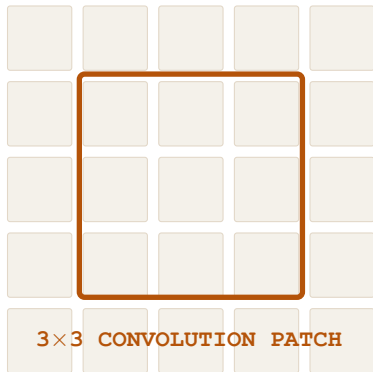
What if the **connections** are?



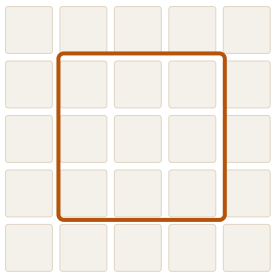
Wait... why is everyone taking orders from this guy?



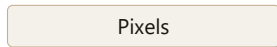




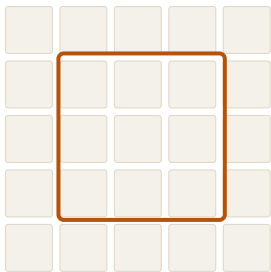
LOCAL CONNECTIVITY · FIXED SPATIAL GRID



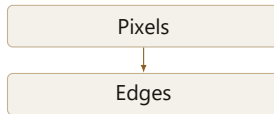
LOCAL PATCHES

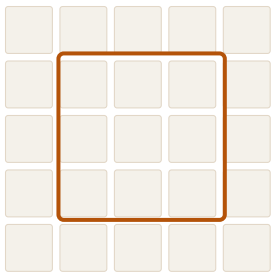


Pixels

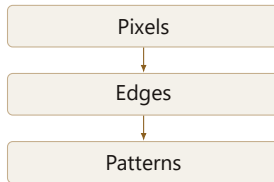


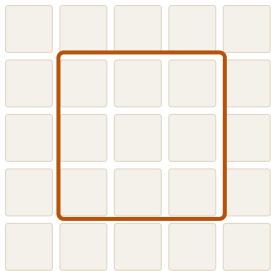
LOCAL PATCHES



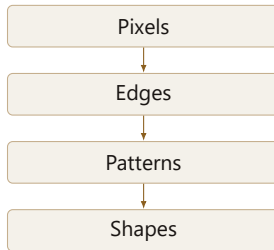


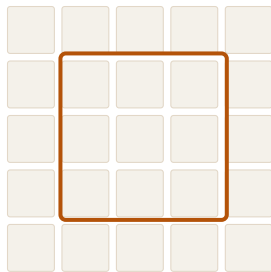
LOCAL PATCHES



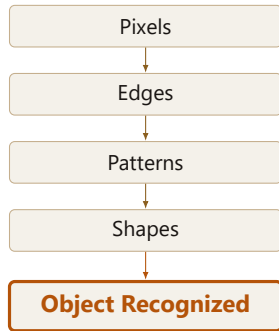


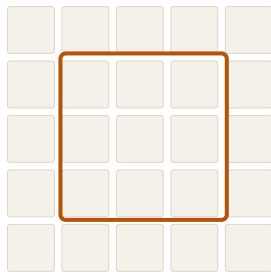
LOCAL PATCHES



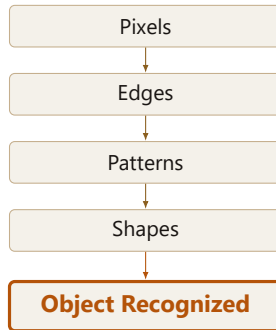


REGULAR PIXEL GRID





REGULAR PIXEL GRID



RIGID GEOMETRIC HIERARCHY

Convolutional Neural Network

Convolutional Neural Network

CNN

Convolutional Neural Network

CNN

BUILT FOR REGULAR EUCLIDEAN GRIDS

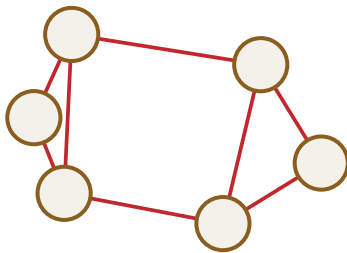




What happens when the
data has no grid?



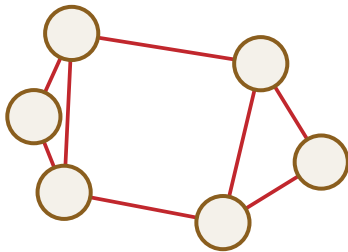
REGULAR GRID



IRREGULAR GRAPH



REGULAR GRID

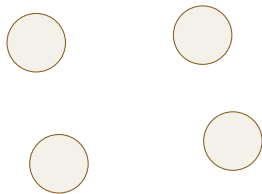


IRREGULAR GRAPH

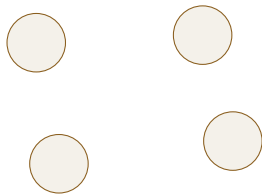
There is no pixel 37.

NO FIXED ORDER · ARBITRARY NEIGHBORS

**SOLUTION 1:
DISCARD THE EDGES**



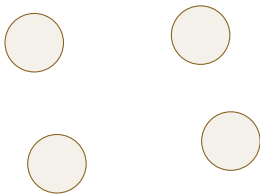
**SOLUTION 1:
DISCARD THE EDGES**



LOSES GRAPH STRUCTURE

What if we keep them?

SOLUTION 1:
DISCARD THE EDGES



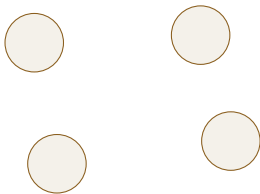
LOSES GRAPH STRUCTURE

SOLUTION 2:
FLATTEN THE ADJACENCY MATRIX

$A \rightarrow [0, 1, 0, 1, 1, 0, 1, 0, \dots]$

$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

SOLUTION 1:
DISCARD THE EDGES



LOSES GRAPH STRUCTURE

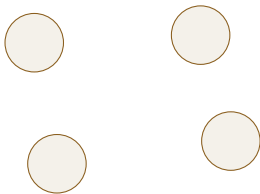
SOLUTION 2:
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SENSITIVE TO NODE ORDERING

SOLUTION 1:
DISCARD THE EDGES



LOSES GRAPH STRUCTURE

SOLUTION 2:
FLATTEN THE ADJACENCY MATRIX

$A \rightarrow [0, 1, 0, 1, 1, 0, 1, 0, \dots]$

$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

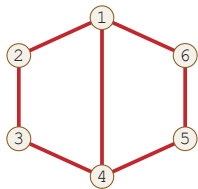
SENSITIVE TO NODE ORDERING

Why do naive approaches fail?

TASK: PREDICT WHETHER THIS MOLECULE IS TOXIC

Same molecule. Same atoms. Same bonds.

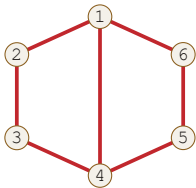
NUMBERING A



TASK: PREDICT WHETHER THIS MOLECULE IS TOXIC

Same molecule. Same atoms. Same bonds.

NUMBERING A



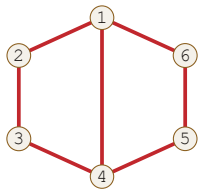
Toxicity: 0.83

ILLUSTRATIVE PREDICTION

TASK: PREDICT WHETHER THIS MOLECULE IS TOXIC

Same molecule. Same atoms. Same bonds.

NUMBERING A



Toxicity: 0.83

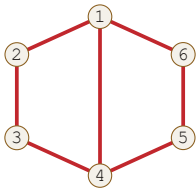
ILLUSTRATIVE PREDICTION

What if we simply
renumber the atoms?

TASK: PREDICT WHETHER THIS MOLECULE IS TOXIC

Same molecule. Same atoms. Same bonds.

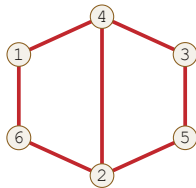
NUMBERING A



Toxicity: 0.83

ILLUSTRATIVE PREDICTION

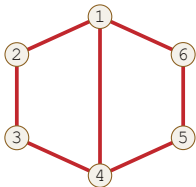
NUMBERING B



TASK: PREDICT WHETHER THIS MOLECULE IS TOXIC

Same molecule. Same atoms. Same bonds.

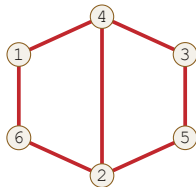
NUMBERING A



Toxicity: 0.83

ILLUSTRATIVE PREDICTION

NUMBERING B



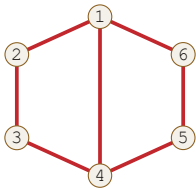
Toxicity: 0.11

ILLUSTRATIVE PREDICTION

TASK: PREDICT WHETHER THIS MOLECULE IS TOXIC

Same molecule. Same atoms. Same bonds.

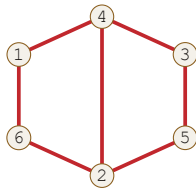
NUMBERING A



Toxicity: 0.83

ILLUSTRATIVE PREDICTION

NUMBERING B



Toxicity: 0.11

ILLUSTRATIVE PREDICTION

PERMUTATION SENSITIVITY · ARBITRARY NUMBERING BIAS

NUMBERING A

Toxicity: 0.83

NUMBERING B

Toxicity: 0.11

NUMBERING A

Toxicity: 0.83

NUMBERING B

Toxicity: 0.11

NUMBERING A

Toxicity: 0.83

NUMBERING B

Toxicity: 0.11

The network learned our filing system.

NUMBERING A

Toxicity: 0.83

NUMBERING B

Toxicity: 0.11

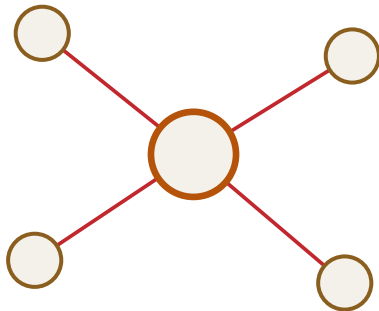
**The network learned our filing system.
It never learned the chemistry.**

What is this node?



What **is** this node?

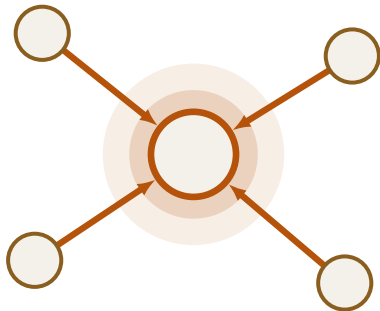
What **is** connected to it?



What **is** this node?

What is connected to it?

Who influences this node?

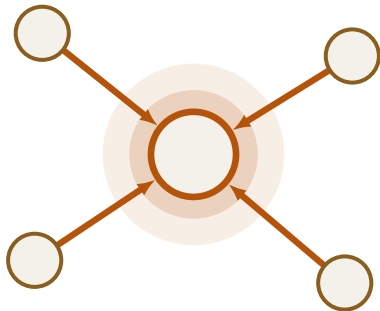


What **is** this node?

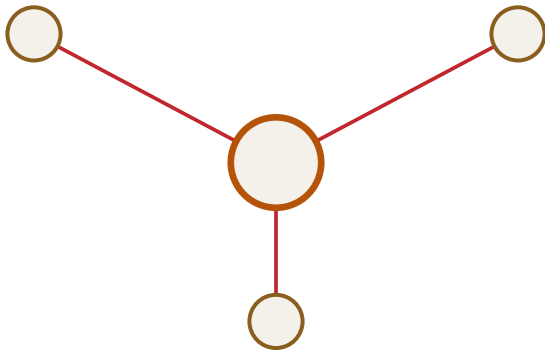
What is connected to it?

Who influences this node?

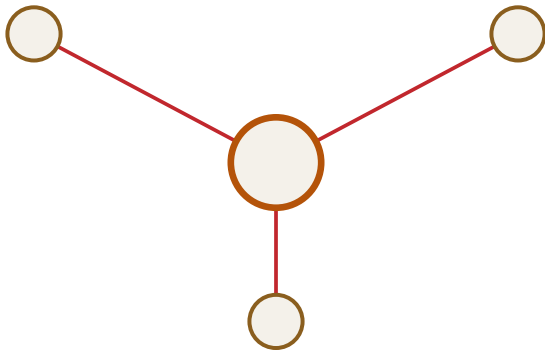
RELATIONAL REASONING



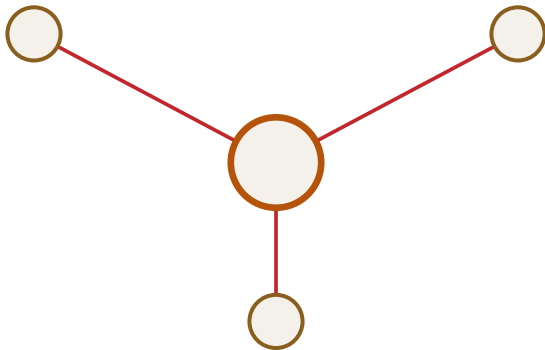
We need to use the neighbourhood.



That is the idea behind a **Graph Neural Network.**

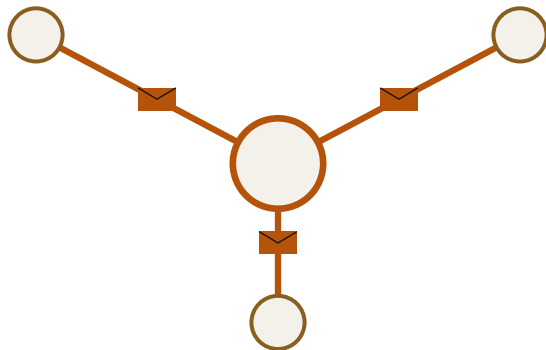


It passes messages between connected nodes.



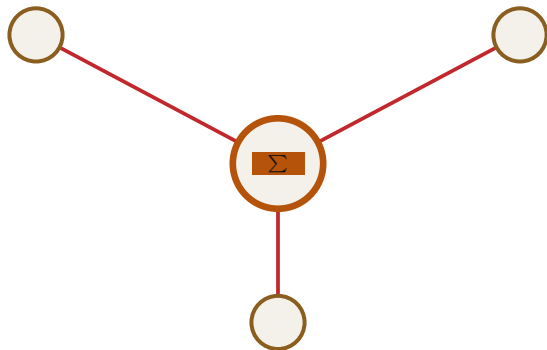
MESSAGE PASSING

It passes messages between connected nodes.



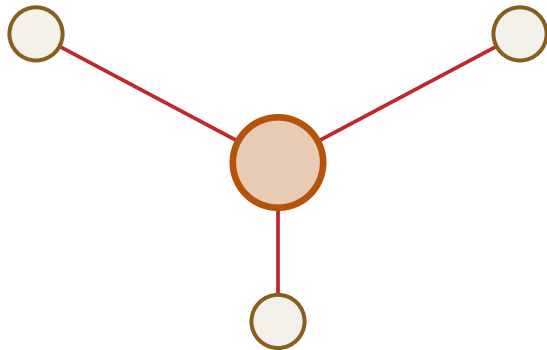
MESSAGE PASSING

It passes messages between connected nodes.



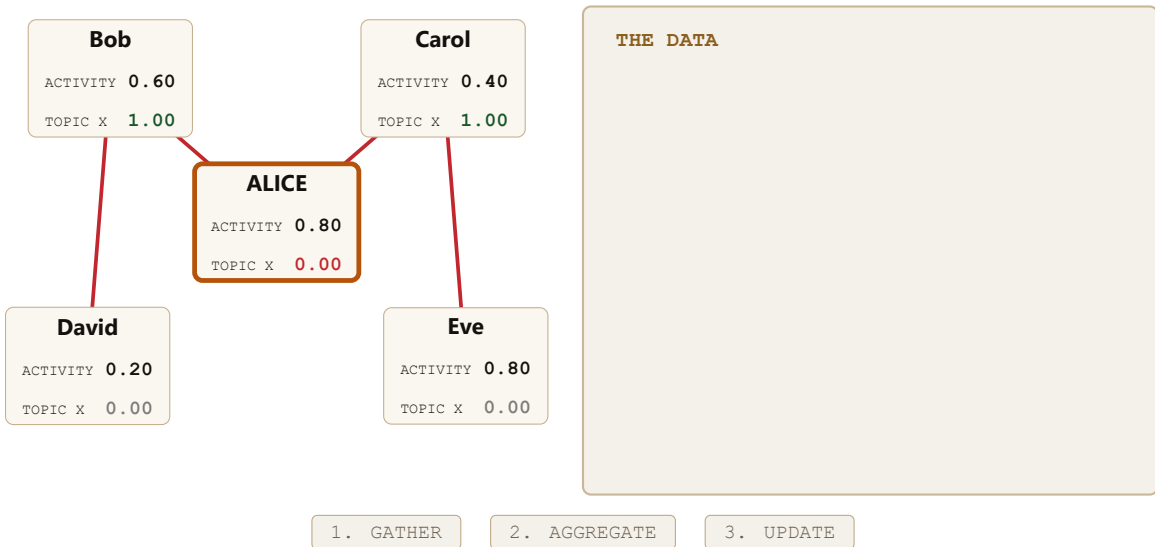
MESSAGE PASSING

It passes messages between connected nodes.

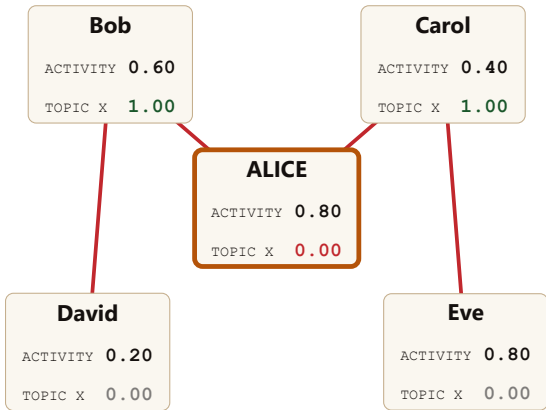


MESSAGE PASSING

Is Alice likely to be interested in Topic X?



Is Alice likely to be interested in Topic X?



THE DATA

SLOT 1 ACTIVITY

how much they post
0 = never 1 = constantly

SLOT 2 TOPIC X

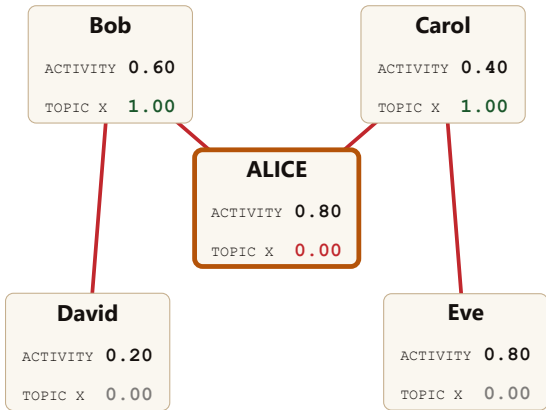
how much they engage with Topic X
0 = never 1 = all the time

1. GATHER

2. AGGREGATE

3. UPDATE

Is Alice likely to be interested in Topic X?



THE DATA

SLOT 1 ACTIVITY

how much they post
0 = never 1 = constantly

SLOT 2 TOPIC X

how much they engage with Topic X
0 = never 1 = all the time

THE FIVE CARDS

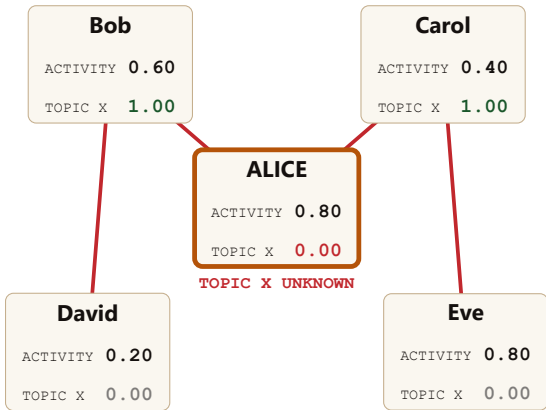
Bob [0.60 , 1.00]
Carol [0.40 , 1.00]
David [0.20 , 0.00]
Eve [0.80 , 0.00]
Alice [0.80 , 0.00]

1. GATHER

2. AGGREGATE

3. UPDATE

Is Alice likely to be interested in Topic X?



THE DATA

SLOT 1 ACTIVITY

how much they post
0 = never 1 = constantly

SLOT 2 TOPIC X

how much they engage with Topic X
0 = never 1 = all the time

THE FIVE CARDS

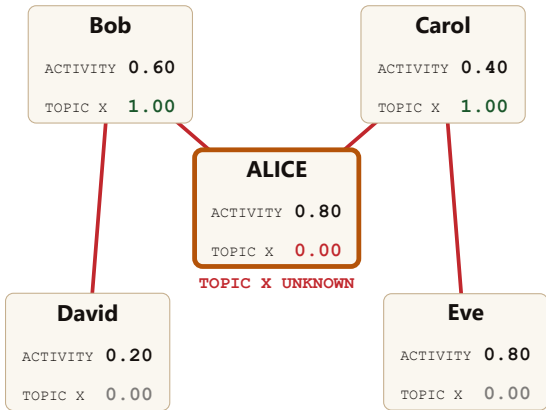
Bob [0.60 , 1.00]
Carol [0.40 , 1.00]
David [0.20 , 0.00]
Eve [0.80 , 0.00]
Alice [0.80 , 0.00]

1. GATHER

2. AGGREGATE

3. UPDATE

Is Alice likely to be interested in Topic X?



THE DATA

SLOT 1 ACTIVITY

how much they post
0 = never 1 = constantly

SLOT 2 TOPIC X

how much they engage with Topic X
0 = never 1 = all the time

THE FIVE CARDS

Bob	[0.60 , 1.00]
Carol	[0.40 , 1.00]
David	[0.20 , 0.00]
Eve	[0.80 , 0.00]
Alice	[0.80 , 0.00]

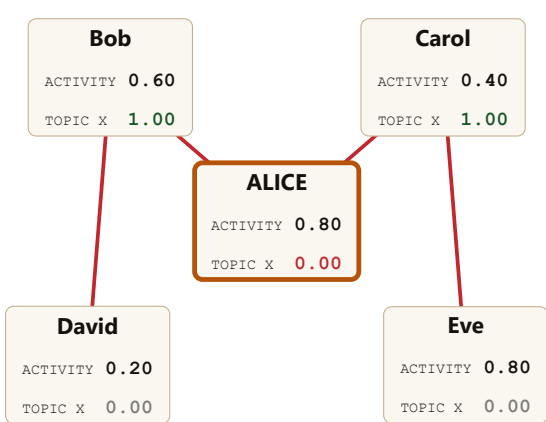
1. GATHER

2. AGGREGATE

3. UPDATE

= ONE LAYER

Step 1 of 3 — GATHER



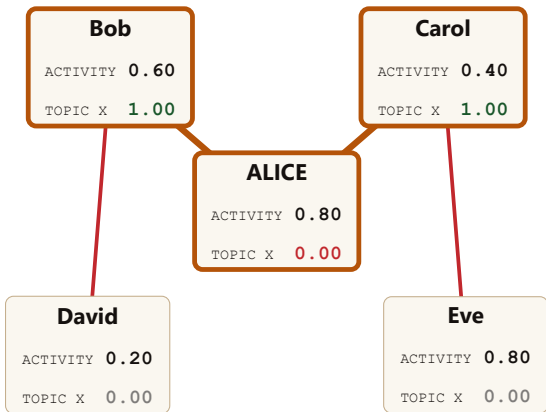
STEP 1 GATHER

1. GATHER

2. AGGREGATE

3. UPDATE

Step 1 of 3 — GATHER



STEP 1 GATHER

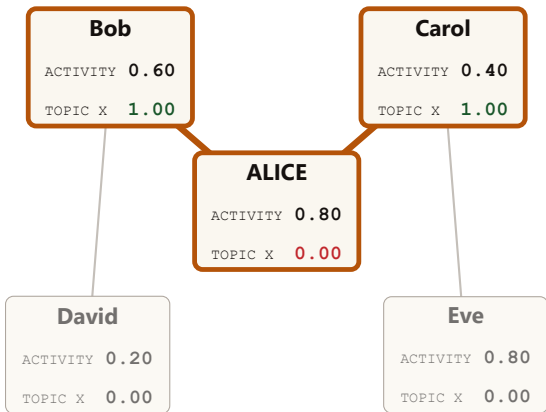
$N(\text{Alice}) = \{ \text{Bob}, \text{Carol} \}$

1. GATHER

2. AGGREGATE

3. UPDATE

Step 1 of 3 — GATHER



STEP 1 GATHER

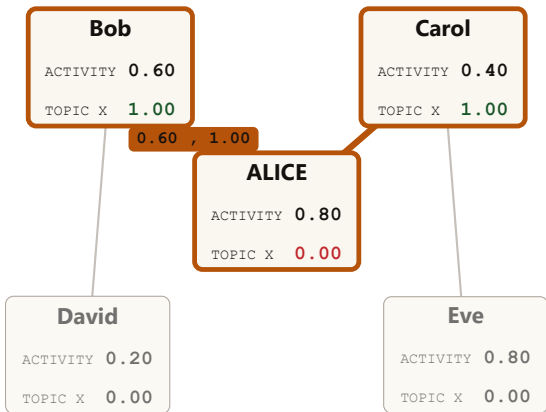
$N(\text{Alice}) = \{ \text{Bob}, \text{Carol} \}$

1. GATHER

2. AGGREGATE

3. UPDATE

Step 1 of 3 — GATHER



STEP 1 GATHER

$N(\text{Alice}) = \{ \text{Bob}, \text{Carol} \}$

WHAT IS A MESSAGE?

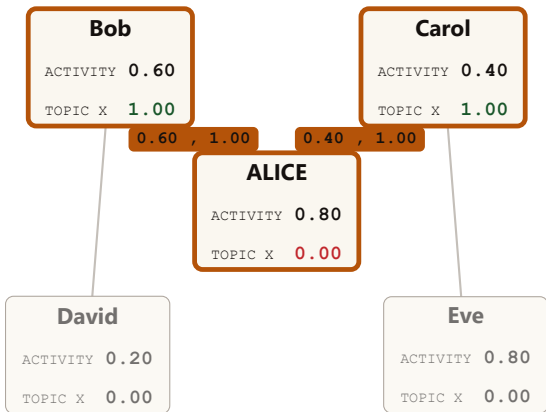
Bob sends [0.60 , 1.00]

1. GATHER

2. AGGREGATE

3. UPDATE

Step 1 of 3 — GATHER



STEP 1 GATHER

$N(\text{Alice}) = \{ \text{Bob} , \text{Carol} \}$

WHAT IS A MESSAGE?

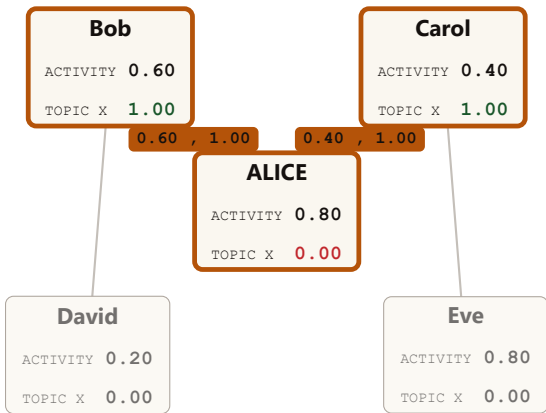
Bob sends [0.60 , 1.00]
Carol sends [0.40 , 1.00]

1. GATHER

2. AGGREGATE

3. UPDATE

Step 1 of 3 — GATHER



STEP 1 GATHER

$N(\text{Alice}) = \{ \text{Bob} , \text{Carol} \}$

WHAT IS A MESSAGE?

Bob sends [0.60 , 1.00]
Carol sends [0.40 , 1.00]

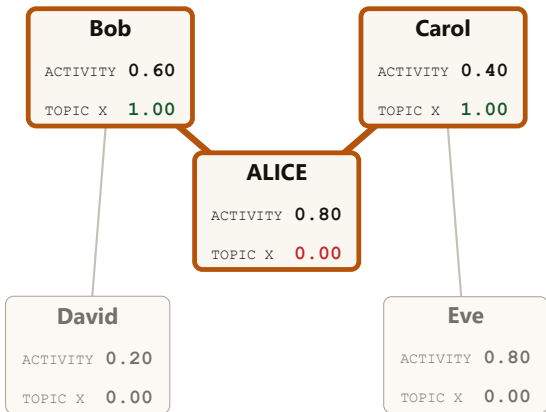
Gathering = Copying messages

1. GATHER

2. AGGREGATE

3. UPDATE

Step 2 of 3 — AGGREGATE



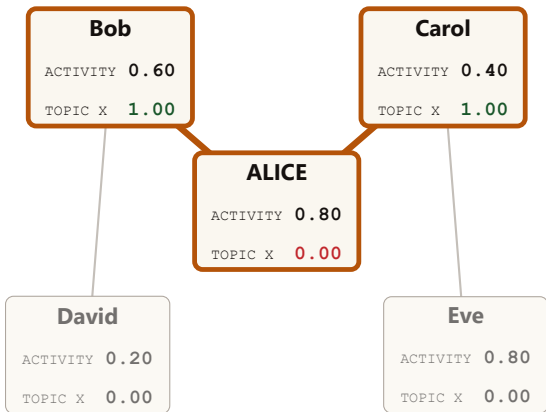
STEP 2 AGGREGATE

1. GATHER

2. AGGREGATE

3. UPDATE

Step 2 of 3 — AGGREGATE



STEP 2 AGGREGATE

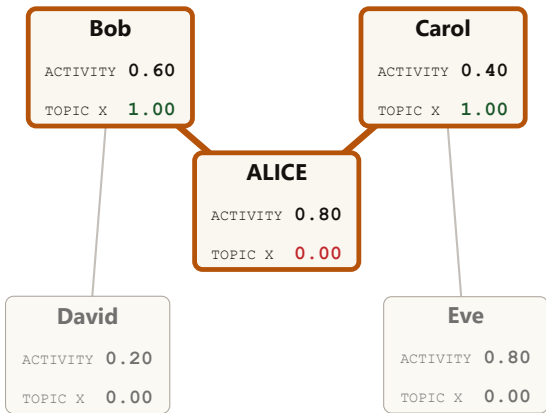
```
Bob    [ 0.60 , 1.00 ]  
Carol  [ 0.40 , 1.00 ]
```

1. GATHER

2. AGGREGATE

3. UPDATE

Step 2 of 3 — AGGREGATE



STEP 2 AGGREGATE

```
Bob    [ 0.60 , 1.00 ]  
Carol  [ 0.40 , 1.00 ]
```

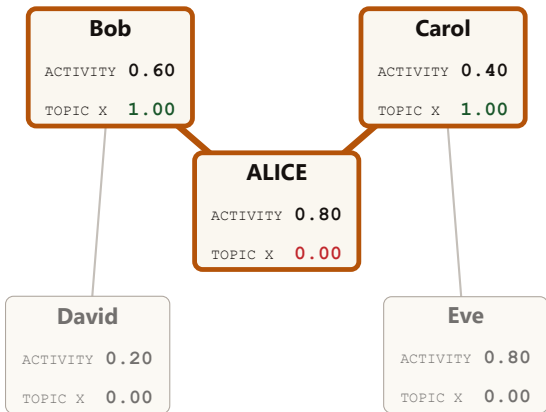
Rule: average them, slot by slot.

1. GATHER

2. AGGREGATE

3. UPDATE

Step 2 of 3 — AGGREGATE



STEP 2 AGGREGATE

```
Bob    [ 0.60 , 1.00 ]  
Carol  [ 0.40 , 1.00 ]
```

Rule: average them, slot by slot.

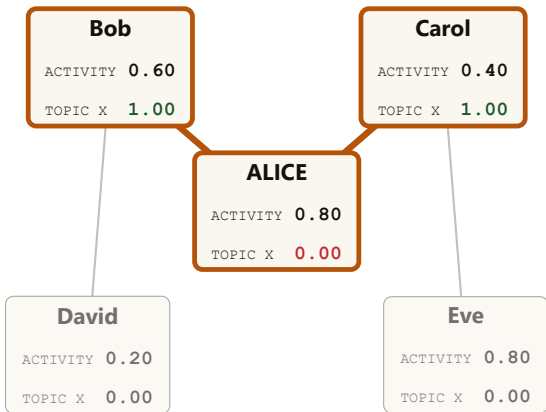
```
ACTIVITY (0.60 + 0.40) / 2 = 0.50
```

1. GATHER

2. AGGREGATE

3. UPDATE

Step 2 of 3 — AGGREGATE



STEP 2 AGGREGATE

Bob [0.60 , 1.00]
Carol [0.40 , 1.00]

Rule: average them, slot by slot.

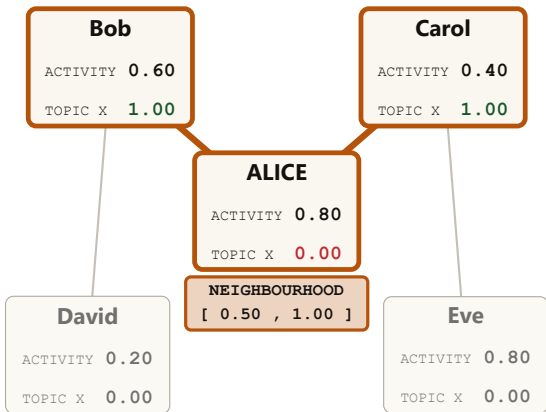
ACTIVITY $(0.60 + 0.40) / 2 = 0.50$
TOPIC X $(1.00 + 1.00) / 2 = 1.00$

1. GATHER

2. AGGREGATE

3. UPDATE

Step 2 of 3 — AGGREGATE



STEP 2 AGGREGATE

```
Bob    [ 0.60 , 1.00 ]  
Carol  [ 0.40 , 1.00 ]
```

Rule: average them, slot by slot.

```
ACTIVITY (0.60 + 0.40) / 2 = 0.50  
TOPIC X  (1.00 + 1.00) / 2 = 1.00
```

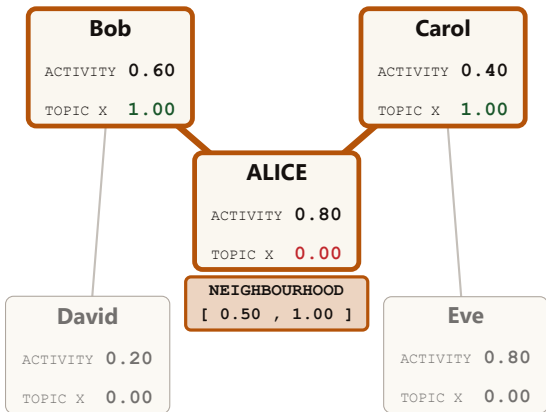
m(Alice) = [0.50 , 1.00]

1. GATHER

2. AGGREGATE

3. UPDATE

Step 2 of 3 — AGGREGATE



STEP 2 AGGREGATE

```
Bob    [ 0.60 , 1.00 ]  
Carol  [ 0.40 , 1.00 ]
```

Rule: average them, slot by slot.

```
ACTIVITY (0.60 + 0.40) / 2 = 0.50  
TOPIC X  (1.00 + 1.00) / 2 = 1.00
```

m(Alice) = [0.50 , 1.00]

ORDER INVARIANCE

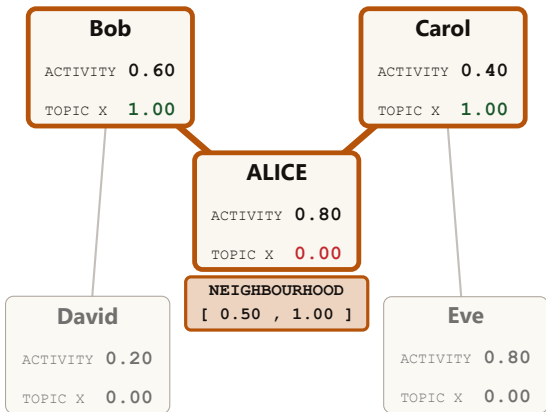
mean(Bob, Carol) = mean(Carol, Bob) = 0.50

1. GATHER

2. AGGREGATE

3. UPDATE

Step 2 of 3 — AGGREGATE



STEP 2 AGGREGATE

```
Bob    [ 0.60 , 1.00 ]  
Carol  [ 0.40 , 1.00 ]
```

Rule: average them, slot by slot.

```
ACTIVITY (0.60 + 0.40) / 2 = 0.50  
TOPIC X  (1.00 + 1.00) / 2 = 1.00
```

m(Alice) = [0.50 , 1.00]

ORDER INVARIANCE

mean(Bob, Carol) = mean(Carol, Bob) = 0.50

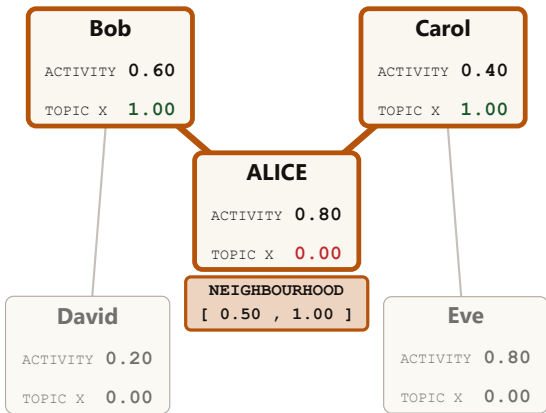
Order Invariant Aggregation

1. GATHER

2. AGGREGATE

3. UPDATE

Step 3 of 3 — UPDATE



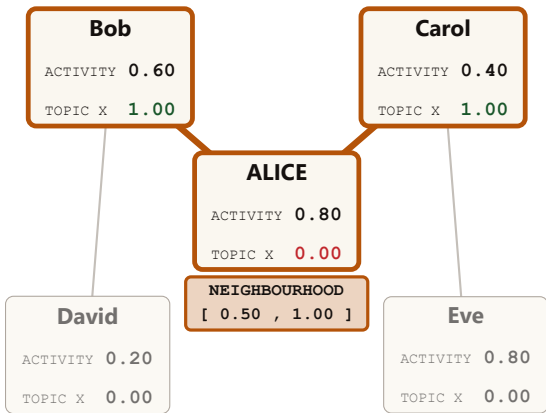
STEP 3 UPDATE

1. GATHER

2. AGGREGATE

3. UPDATE

Step 3 of 3 — UPDATE



STEP 3 UPDATE

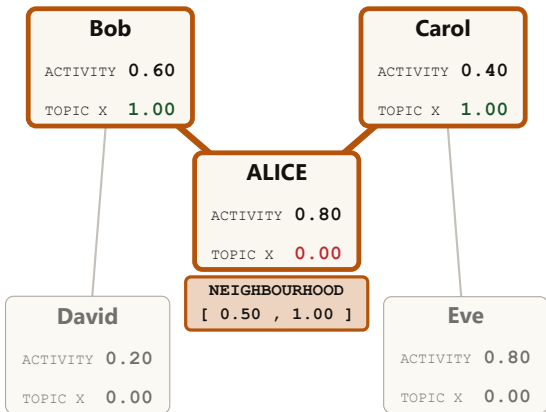
```
mine    [ 0.80 , 0.00 ]  
theirs  [ 0.50 , 1.00 ]
```

1. GATHER

2. AGGREGATE

3. UPDATE

Step 3 of 3 — UPDATE



STEP 3 UPDATE

```
mine    [ 0.80 , 0.00 ]  
theirs  [ 0.50 , 1.00 ]
```

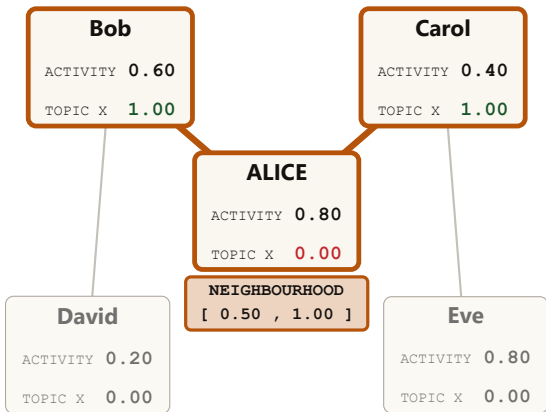
new = 0.5 x mine + 0.5 x theirs

1. GATHER

2. AGGREGATE

3. UPDATE

Step 3 of 3 — UPDATE



STEP 3 UPDATE

```
mine    [ 0.80 , 0.00 ]  
theirs  [ 0.50 , 1.00 ]
```

new = 0.5 x mine + 0.5 x theirs

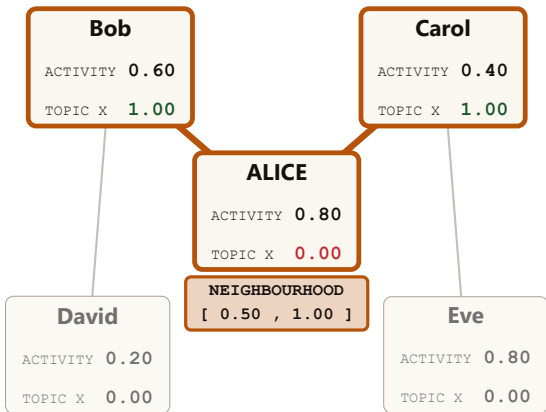
```
ACTIVITY 0.5(0.80) + 0.5(0.50)  
          = 0.40 + 0.25 = 0.65
```

1. GATHER

2. AGGREGATE

3. UPDATE

Step 3 of 3 — UPDATE



STEP 3 UPDATE

```
mine    [ 0.80 , 0.00 ]  
theirs  [ 0.50 , 1.00 ]
```

new = 0.5 x mine + 0.5 x theirs

```
ACTIVITY  0.5(0.80) + 0.5(0.50)  
           = 0.40 + 0.25 = 0.65
```

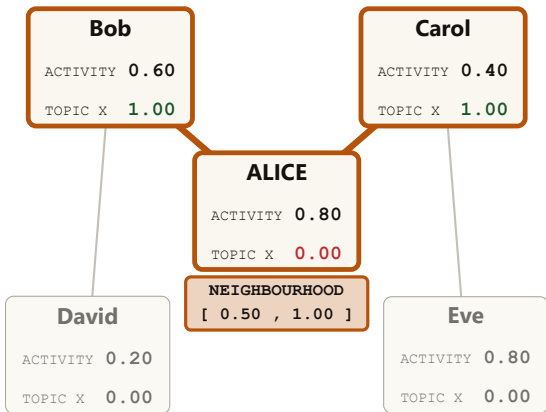
```
TOPIC X   0.5(0.00) + 0.5(1.00)  
           = 0.00 + 0.50 = 0.50
```

1. GATHER

2. AGGREGATE

3. UPDATE

Step 3 of 3 — UPDATE



STEP 3 UPDATE

```
mine    [ 0.80 , 0.00 ]  
theirs  [ 0.50 , 1.00 ]
```

new = 0.5 x mine + 0.5 x theirs

```
ACTIVITY  0.5(0.80) + 0.5(0.50)  
           = 0.40 + 0.25 = 0.65
```

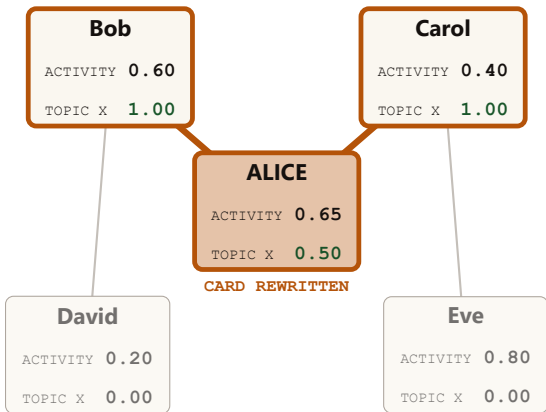
```
TOPIC X   0.5(0.00) + 0.5(1.00)  
           = 0.00 + 0.50 = 0.50
```

1. GATHER

2. AGGREGATE

3. UPDATE

Step 3 of 3 — UPDATE



STEP 3 UPDATE

```
mine    [ 0.80 , 0.00 ]  
theirs  [ 0.50 , 1.00 ]
```

new = 0.5 x mine + 0.5 x theirs

```
ACTIVITY  0.5(0.80) + 0.5(0.50)  
           = 0.40 + 0.25 = 0.65
```

```
TOPIC X    0.5(0.00) + 0.5(1.00)  
           = 0.00 + 0.50 = 0.50
```

h'(Alice) = [0.65 , 0.50]

1. GATHER

2. AGGREGATE

3. UPDATE

One blank slot, filled in by the neighbourhood.

ALICE

ACTIVITY 0.80

TOPIC X 0.00

One blank slot, filled in by the neighbourhood.

ALICE	
ACTIVITY	0.80
TOPIC X	0.00



ALICE	
ACTIVITY	0.65
TOPIC X	0.50

TOPIC X: 0.00 → 0.50

One blank slot, filled in by the neighbourhood.

ALICE	
ACTIVITY	0.80
TOPIC X	0.00



ALICE	
ACTIVITY	0.65
TOPIC X	0.50

TOPIC X: 0.00 → 0.50

READOUT LAYER

One blank slot, filled in by the neighbourhood.

ALICE	
ACTIVITY	0.80
TOPIC X	0.00



ALICE	
ACTIVITY	0.65
TOPIC X	0.50

TOPIC X: 0.00 → 0.50

READOUT LAYER

$$p = \sigma(6 \times 0.50 - 0.9) = \sigma(2.1) = 0.89$$

One blank slot, filled in by the neighbourhood.

ALICE	
ACTIVITY	0.80
TOPIC X	0.00



ALICE	
ACTIVITY	0.65
TOPIC X	0.50

TOPIC X: 0.00 → 0.50

READOUT LAYER

$$p = \sigma(6 \times 0.50 - 0.9) = \sigma(2.1) = 0.89$$

LIKELY INTERESTED 89%

One blank slot, filled in by the neighbourhood.

ALICE	
ACTIVITY	0.80
TOPIC X	0.00



ALICE	
ACTIVITY	0.65
TOPIC X	0.50

TOPIC X: 0.00 → 0.50

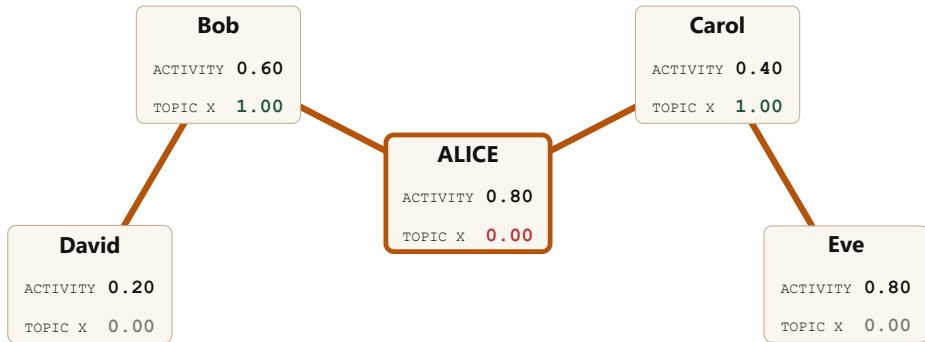
READOUT LAYER

$$p = \sigma(6 \times 0.50 - 0.9) = \sigma(2.1) = 0.89$$

LIKELY INTERESTED 89%

PREDICTION DERIVED FROM CONNECTIONS

Alice was not special. Every node ran the same three steps.



Alice was not special. Every node ran the same three steps.



Alice was not special. Every node ran the same three steps.



Alice was not special. Every node ran the same three steps.



ALL NODES CONVERGE TOWARD 0.50 IN SLOT 2

Alice was not special. Every node ran the same three steps.



ALL NODES CONVERGE TOWARD 0.50 IN SLOT 2

POTENTIAL BOTTLENECK: OVER-SMOOTHING

$$h_v^{(k+1)} = \sigma \left(W_{\text{self}} h_v^{(k)} + W_{\text{nbr}} \text{AGG} \left(\{ h_u^{(k)} : u \in \mathcal{N}(v) \} \right) \right)$$

$$h_v^{(k+1)} = \sigma \left(W_{\text{self}} h_v^{(k)} + W_{\text{nbr}} \text{AGG} \left(\{ h_u^{(k)} : u \in \mathcal{N}(v) \} \right) \right)$$

my neighbours
{ Bob , Carol }



$$h_v^{(k+1)} = \sigma \left(W_{\text{self}} h_v^{(k)} + W_{\text{nbr}} \text{AGG} \left(\{ h_u^{(k)} : u \in \mathcal{N}(v) \} \right) \right)$$

each neighbour's current card

Bob = [0.60 , 1.00]

my neighbours

{ Bob , Carol }

$$h_v^{(k+1)} = \sigma \left(W_{\text{self}} h_v^{(k)} + W_{\text{nbr}} \text{AGG} \left(\{ h_u^{(k)} : u \in \mathcal{N}(v) \} \right) \right)$$

the shuffle-proof combiner
average = [0.50 , 1.00]

each neighbour's current card
Bob = [0.60 , 1.00]

my neighbours
{ Bob , Carol }

The diagram illustrates the equation for the next hidden state $h_v^{(k+1)}$ of a node v . The equation is:

$$h_v^{(k+1)} = \sigma \left(W_{\text{self}} h_v^{(k)} + W_{\text{nbr}} \text{AGG} \left(\{ h_u^{(k)} : u \in \mathcal{N}(v) \} \right) \right)$$

Annotations explaining the components:

- the shuffle-proof combiner**
average = [0.50 , 1.00]
- how much of them
0.5
- each neighbour's current card
Bob = [0.60 , 1.00]
- my neighbours
{ Bob , Carol }

Diagram illustrating the equation for $h_v^{(k+1)}$ with annotations:

Annotations:

- how much of me
 $0.5 \times [0.80, 0.00]$
- how much of them
0.5
- the shuffle-proof combiner
average = $[0.50, 1.00]$
- each neighbour's current card
Bob = $[0.60, 1.00]$
- my neighbours
 $\{ \text{Bob}, \text{Carol} \}$

Equation:

$$h_v^{(k+1)} = \sigma \left(W_{\text{self}} h_v^{(k)} + W_{\text{nbr}} \text{AGG} \left(\{ h_u^{(k)} : u \in \mathcal{N}(v) \} \right) \right)$$

how much of me
 $0.5 \times [0.80, 0.00]$

the shuffle-proof combiner
average = $[0.50, 1.00]$

how much of them
0.5

squash it back
into range

$$h_v^{(k+1)} = \sigma \left(W_{\text{self}} h_v^{(k)} + W_{\text{nbr}} \text{AGG} \left(\{ h_u^{(k)} : u \in \mathcal{N}(v) \} \right) \right)$$

each neighbour's current card
Bob = $[0.60, 1.00]$

my neighbours
 $\{ \text{Bob}, \text{Carol} \}$

The diagram illustrates the components of the equation for $h_v^{(k+1)}$. The equation is $h_v^{(k+1)} = \sigma \left(W_{\text{self}} h_v^{(k)} + W_{\text{nbr}} \text{AGG} \left(\{ h_u^{(k)} : u \in \mathcal{N}(v) \} \right) \right)$. Annotations include: 'squash it back into range' pointing to the σ function; 'how much of me' with value $0.5 \times [0.80, 0.00]$ pointing to W_{self} ; 'the shuffle-proof combiner' with value 'average = $[0.50, 1.00]$ ' pointing to the AGG function; 'how much of them' with value 0.5 pointing to W_{nbr} ; 'each neighbour's current card' with example 'Bob = $[0.60, 1.00]$ ' pointing to the set $\{ h_u^{(k)} \}$; and 'my neighbours' with example $\{ \text{Bob}, \text{Carol} \}$ pointing to the set definition $u \in \mathcal{N}(v)$.

Diagram illustrating the equation for $h_v^{(k+1)}$ and its components:

$$h_v^{(k+1)} = \sigma \left(W_{\text{self}} h_v^{(k)} + W_{\text{nbr}} \text{AGG} \left(\{ h_u^{(k)} : u \in \mathcal{N}(v) \} \right) \right)$$

Annotations and their corresponding parts in the equation:

- how much of me**
 $0.5 \times [0.80 , 0.00]$
Points to $W_{\text{self}} h_v^{(k)}$
- how much of them**
 0.5
Points to W_{nbr}
- the shuffle-proof combiner**
average = $[0.50 , 1.00]$
Points to AGG
- each neighbour's current card**
Bob = $[0.60 , 1.00]$
Points to $\{ h_u^{(k)} : u \in \mathcal{N}(v) \}$
- my neighbours**
 $\{ \text{Bob} , \text{Carol} \}$
Points to $\mathcal{N}(v)$
- squash it back into range**
Points to σ
- Alice's new card**
 $[0.65 , 0.50]$
Points to $h_v^{(k+1)}$

how much of me
 $0.5 \times [0.80, 0.00]$

the shuffle-proof combiner
 average = $[0.50, 1.00]$

how much of them
 0.5

squash it back into range

Alice's new card
 $[0.65, 0.50]$

each neighbour's current card
 Bob = $[0.60, 1.00]$

my neighbours
 $\{ \text{Bob}, \text{Carol} \}$

$$h_v^{(k+1)} = \sigma \left(W_{\text{self}} h_v^{(k)} + W_{\text{nbr}} \text{AGG} \left(\{ h_u^{(k)} : u \in \mathcal{N}(v) \} \right) \right)$$

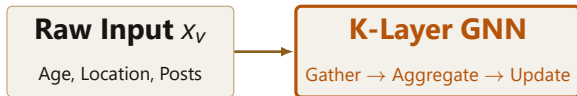
Gather · Aggregate · Update

From Message Passing to Predictions

Raw Input X_V

Age, Location, Posts

From Message Passing to Predictions



From Message Passing to Predictions



From Message Passing to Predictions



k-HOP NEIGHBORHOOD EMBEDDINGS

Scenario: Classifying the Dhaka Suspect

SUSPECT EMBEDDING h_{target}

Target #4091

Scenario: Classifying the Dhaka Suspect

SUSPECT EMBEDDING h_{target}

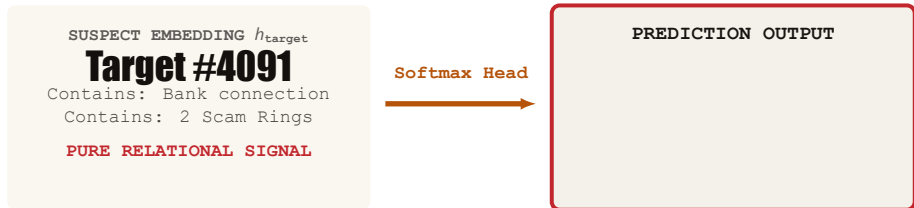
Target #4091

Contains: Bank connection

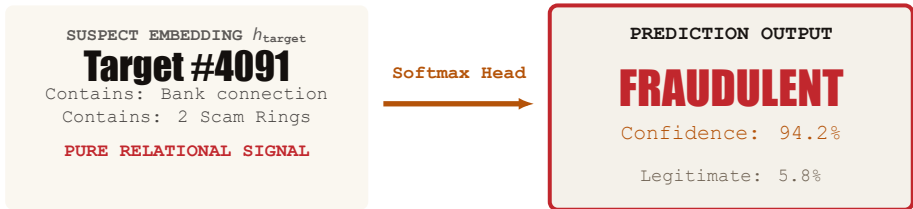
Contains: 2 Scam Rings

PURE RELATIONAL SIGNAL

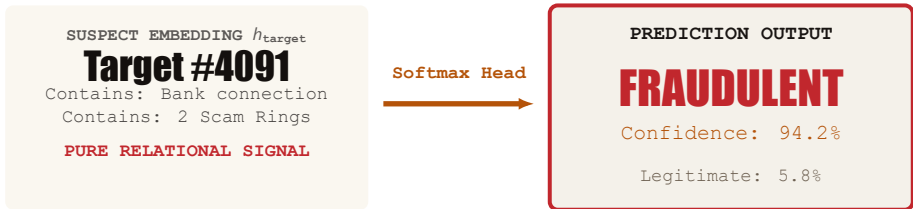
Scenario: Classifying the Dhaka Suspect



Scenario: Classifying the Dhaka Suspect



Scenario: Classifying the Dhaka Suspect



GUILT BY ASSOCIATION · LEARNED MATHEMATICALLY

NODE LEVEL

Classification

Is this account
a scammer?

$$\hat{y}_v = \text{Softmax}(Wh_v)$$

THE THREE GRAPH PREDICTION TASKS

NODE LEVEL

Classification

Is this account
a scammer?

$$\hat{y}_v = \text{Softmax}(Wh_v)$$

EDGE LEVEL

Link Prediction

Will these two nodes
become connected?

$$\hat{y}_{uv} = \sigma(h_u^T h_v)$$

THE THREE GRAPH PREDICTION TASKS

NODE LEVEL

Classification

Is this account
a scammer?

$$\hat{y}_v = \text{Softmax}(Wh_v)$$

EDGE LEVEL

Link Prediction

Will these two nodes
become connected?

$$\hat{y}_{uv} = \sigma(h_u^T h_v)$$

GRAPH LEVEL

Property Test

Is this entire molecule
toxic or curative?

$$h_G = \text{Pool}(\{h_v\})$$

THE THREE GRAPH PREDICTION TASKS

NODE LEVEL

Classification

Is this account
a scammer?

$$\hat{y}_v = \text{Softmax}(Wh_v)$$

EDGE LEVEL

Link Prediction

Will these two nodes
become connected?

$$\hat{y}_{uv} = \sigma(h_u^T h_v)$$

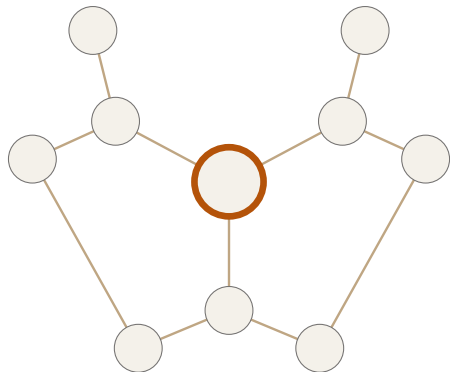
GRAPH LEVEL

Property Test

Is this entire molecule
toxic or curative?

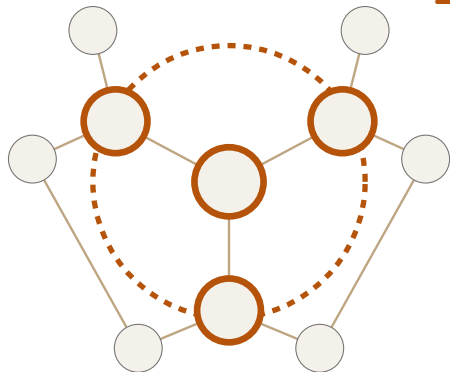
$$h_G = \text{Pool}(\{h_v\})$$

ONE MECHANISM · THREE PREDICTION SCALES

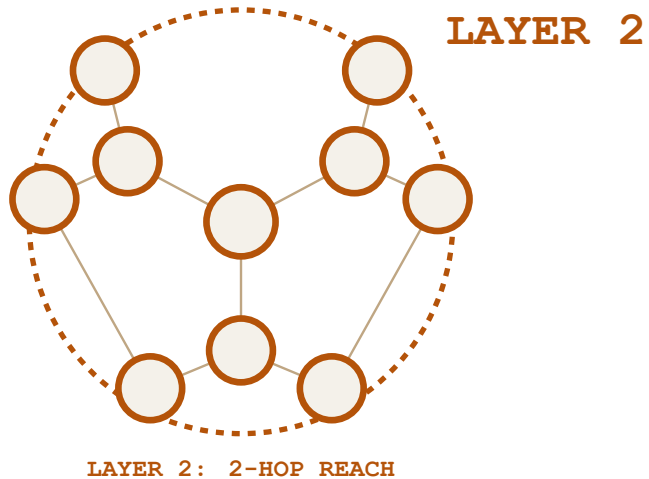


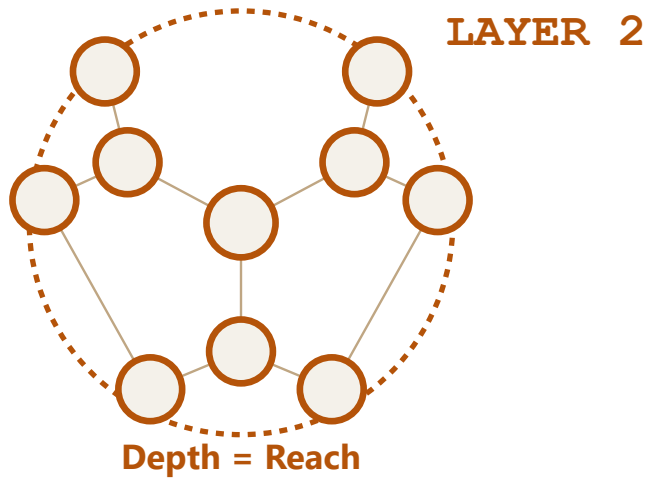
INPUT GRAPH: 1-HOP NEIGHBORHOOD

LAYER 1

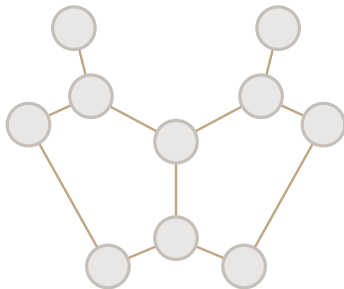


LAYER 1: 1-HOP REACH



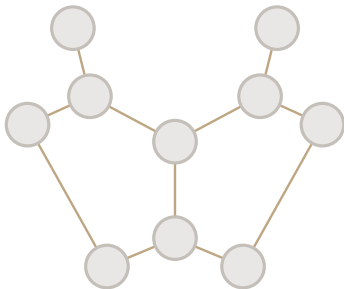


LAYER 6



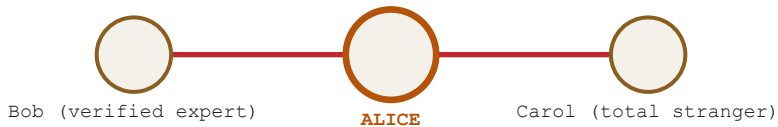
OVER-SMOOTHING

LAYER 6

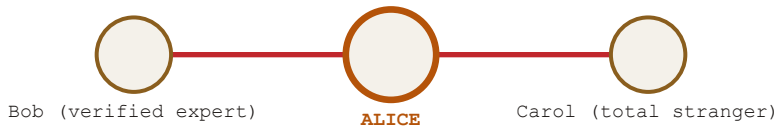


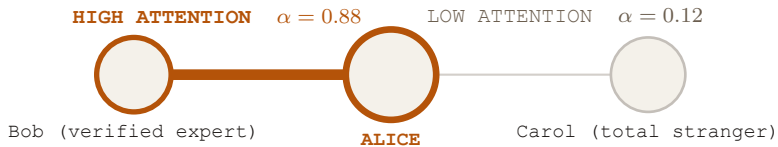
OVER-SMOOTHING
CONVERGENCE TO MEAN · LIMIT: 2-3 LAYERS

We averaged Bob and Carol equally. Should we have?

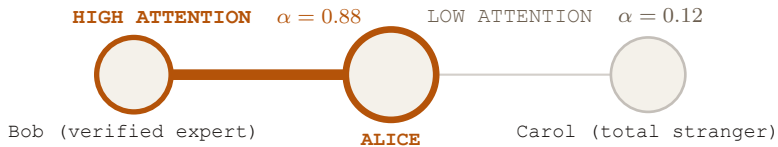


We averaged Bob and Carol equally. Should we have?





LEARNED ATTENTION WEIGHTS (α_{ij})



LEARNED ATTENTION WEIGHTS (α_{ij})

DYNAMIC PER-EDGE ATTENTION COEFFICIENTS

GCN

Average everyone

GCN

Average everyone

GraphSAGE

Sample a few

GCN

Average everyone

GraphSAGE

Sample a few

GAT

Learn who to trust

GCN

Average everyone

GraphSAGE

Sample a few

GAT

Learn who to trust

GATHER · AGGREGATE · ATTEND

GCN

Average everyone

GraphSAGE

Sample a few

GAT

Learn who to trust

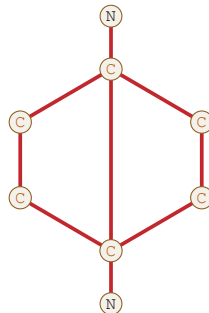
GATHER · AGGREGATE · ATTEND

HOW TO GATHER · HOW TO AGGREGATE · WHO TO LISTEN TO

MIT · 2020

Halicin

HALICIN GRAPH REPRESENTATION

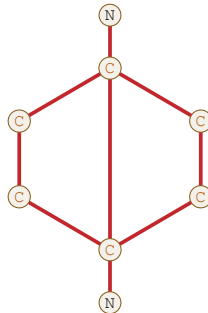


MIT · 2020

Halicin

- Screened **100 million** molecules as graphs.

HALICIN GRAPH REPRESENTATION

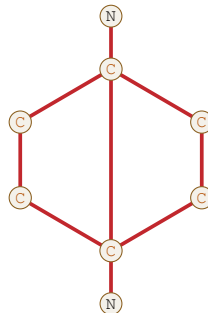


MIT · 2020

Halicin

- Screened **100 million** molecules as graphs.
- Learned chemical bond topologies, not atom numbering.

HALICIN GRAPH REPRESENTATION

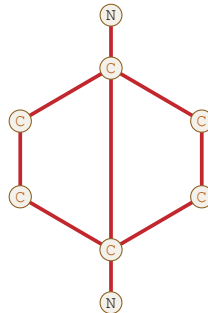


MIT · 2020

Halicin

- Screened **100 million** molecules as graphs.
- Learned chemical bond topologies, not atom numbering.
- Flagged a molecule humans had completely overlooked.

HALICIN GRAPH REPRESENTATION



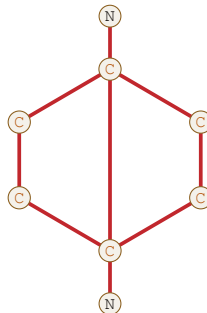
MIT · 2020

Halicin

- Screened **100 million** molecules as graphs.
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- Flagged a molecule humans had completely overlooked.

Killed Pan-Resistant Bacteria

HALICIN GRAPH REPRESENTATION



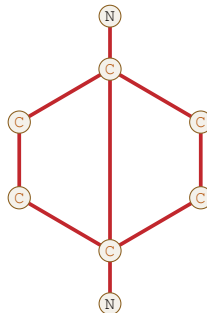
MIT · 2020

Halicin

- Screened **100 million** molecules as graphs.
- Learned chemical bond topologies, not atom numbering.
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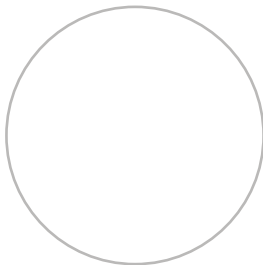
Killed Pan-Resistant Bacteria

HALICIN GRAPH REPRESENTATION

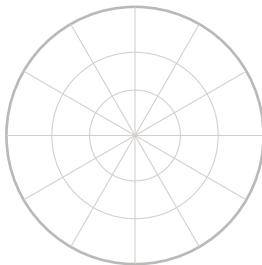


MOLECULAR GRAPH · ANTIBIOTIC DISCOVERY

CASE FILE 02: THE ENTIRE ATMOSPHERE

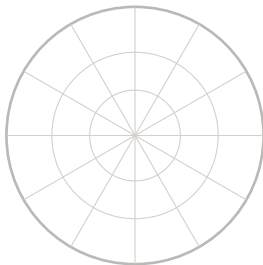


CASE FILE 02: THE ENTIRE ATMOSPHERE



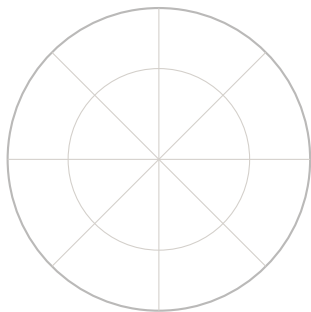
ATMOSPHERIC SPHERICAL MESH GRAPH

CASE FILE 02: THE ENTIRE ATMOSPHERE



ATMOSPHERIC SPHERICAL MESH GRAPH

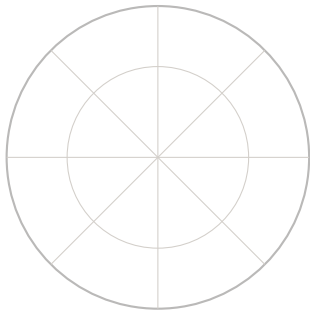
SPHERICAL MESH GRAPH · PLANETARY AIRFLOW



PLANETARY SCALE

DEEPMIND · 2023

GraphCast

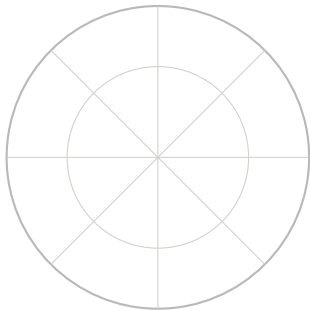


PLANETARY SCALE

DEEPMIND · 2023

GraphCast

- 10-day global forecast in **under 1 minute** on one machine.

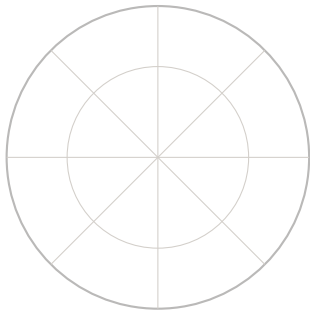


PLANETARY SCALE

DEEPMIND · 2023

GraphCast

- 10-day global forecast in **under 1 minute** on one machine.
- Beat the European gold-standard supercomputers on >90% of variables.

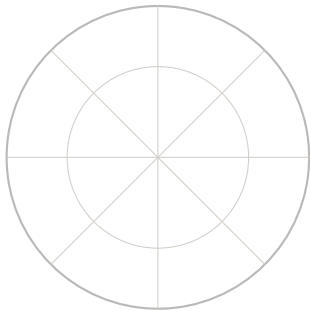


PLANETARY SCALE

DEEPMIND · 2023

GraphCast

- 10-day global forecast in **under 1 minute** on one machine.
- Beat the European gold-standard supercomputers on >90% of variables.
- Predicted Hurricane Lee days earlier.



PLANETARY SCALE

DEEPMIND · 2023

GraphCast

- 10-day global forecast in **under 1 minute** on one machine.
- Beat the European gold-standard supercomputers on >90% of variables.
- Predicted Hurricane Lee days earlier.

Same Principle. Planetary Scale.

CASE FILE 03: CERN LARGE HADRON COLLIDER



Which hits belong to the same particle?

CERN LHC · CMS / ATLAS

Connect Dots



CERN LHC · CMS / ATLAS

Connect Dots

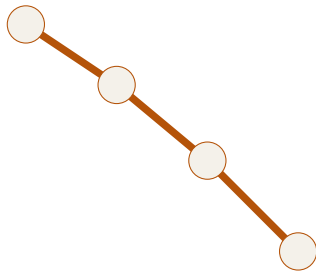
- Builds a graph of candidate hits.



CERN LHC · CMS / ATLAS

Connect Dots

- Builds a graph of candidate hits.
- The GNN prunes false connections in **milliseconds**.

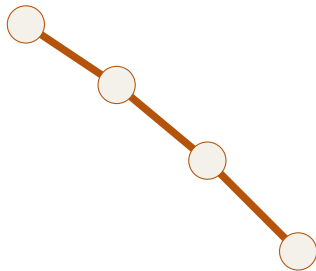


TRUE PARTICLE TRAJECTORY

CERN LHC · CMS / ATLAS

Connect Dots

- Builds a graph of candidate hits.
- The GNN prunes false connections in **milliseconds**.
- Reconstructs particle trajectories in real time.



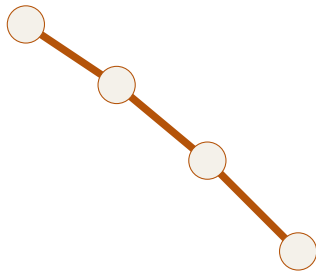
TRUE PARTICLE TRAJECTORY

CERN LHC · CMS / ATLAS

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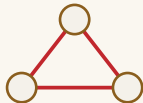
Discovering New Physics



TRUE PARTICLE TRAJECTORY

MOLECULES

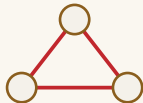
Atoms + Bonds



CHEMISTRY

MOLECULES

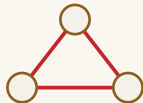
Atoms + Bonds



CHEMISTRY

PLANET

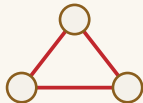
Grid + Airflow



ATMOSPHERE

MOLECULES

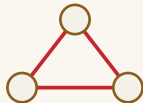
Atoms + Bonds



CHEMISTRY

PLANET

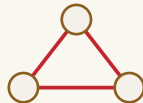
Grid + Airflow



ATMOSPHERE

PARTICLES

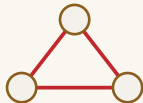
Hits + Tracks



PHYSICS

MOLECULES

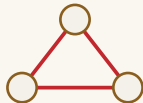
Atoms + Bonds



CHEMISTRY

PLANET

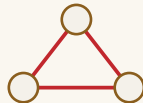
Grid + Airflow



ATMOSPHERE

PARTICLES

Hits + Tracks



PHYSICS

DIVERSE DOMAINS · ONE GRAPH LANGUAGE

GRAPH

GRAPH

Different problems. Same underlying structure.

GRAPH

Different problems. Same underlying structure.

RELATIONSHIPS AS FIRST-CLASS CITIZENS



PROFILE FILE · TARGET #4091

PERSON OF INTEREST

LOCATION: DHAKA, BD AGE: 24

POSTS: 17 STATUS: NORMAL USER

IDENTITY RESOLVED THROUGH CONNECTIONS

VERIFIED BANK



PROFILE FILE · TARGET #4091

PERSON OF INTEREST

LOCATION: DHAKA, BD AGE: 24

POSTS: 17 STATUS: NORMAL USER

IDENTITY RESOLVED THROUGH CONNECTIONS

FLAGGED SCAM OPERATOR

FLAGGED FRAUD RING

VERIFIED BANK



PROFILE FILE · TARGET #4091

PERSON OF INTEREST

LOCATION: DHAKA, BD AGE: 24

POSTS: 17 STATUS: NORMAL USER

IDENTITY RESOLVED THROUGH CONNECTIONS

FLAGGED SCAM OPERATOR

FLAGGED FRAUD RING

The Answer Was in the Connections.

Thank you.

QUESTIONS & DISCUSSION